

Automated Static Characterization Platform for Multi-GaN Devices Across Cryogenic and High-Temperatures

Purushottam Khadka

Electrical, Computer, and Systems Engineering
Rensselaer Polytechnic Institute
Troy, NY, USA
khadkp@rpi.edu

Cheng Wan

Electrical Engineering
Clemson University
Charleston, SC, USA
cwan@clemson.edu

Zheyu Zhang

Electrical, Computer, and Systems Engineering
Rensselaer Polytechnic Institute
Troy, NY, USA
zhangz49@rpi.edu

Tian Qiu

Electrical, Computer, and Systems Engineering
Rensselaer Polytechnic Institute
Troy, NY, USA
qiut2@rpi.edu

Ahmed Siraj

Electrical, Computer, and Systems Engineering
Rensselaer Polytechnic Institute
Troy, NY, USA
siraja@rpi.edu

Abstract—GaN has demonstrated superior characteristics for cryogenic operation and is considered a promising candidate for space power applications. However, much of the characterization work has been conducted using a cryo chamber with LN₂ at 77K, which is insufficient for target space applications such as lunar missions that demand operation down to 50K. Moreover, most studies have provided data based on limited samples, which might not be sufficient for emerging GaN with relatively large tolerance. To address these limitations, an automated characterization platform is proposed. Firstly, the hardware platform, which includes a cryocooler setup, programmable selector, curve tracer, and temperature controller, is discussed. Secondly, the automation workflow, integrated with the MATLAB GUI, for seamless multi-device characterization over a wide temperature range (50K-400K), is described. Thirdly, the platform is validated by characterizing four GaN samples of EPC 2215. The test results reveal (i) a continuous trend of the GaN under test across a wide temperature range of 50K-400K and (ii) data with statistical variations based on three samples, highlighting the need to characterize multiple parts for a full understanding of GaN characteristics.

Index Terms—static characterization, cryogenic temperature, cryocooler, curve tracer, bidirectional switch, relay, automation, programmable selector

I. INTRODUCTION

Due to their unique material properties and construction, Gallium Nitride (GaN) devices have emerged as promising candidates for space applications [1]. In the harsh space environment, electronic devices are exposed to various forms of radiation, which can contribute to system failure. Consequently, space applications require devices that are radiation-resilient, robust, efficient, and lightweight. GaN's intrinsic material

properties make it highly resistant to radiation, significantly enhancing its radiation tolerance compared to silicon (Si) counterparts. GaN transistors demonstrate radiation-hardness performance and single-event effects that are approximately 10 times better than those of Si devices [2]–[4]. Additionally, GaN devices have lower input capacitance and require smaller driving voltages, which improves gate drive system efficiency. Their lower on-state resistance also reduces the need for extensive cooling management, making them lighter and less lossy [5]. GaN systems are utilized in space exploration applications such as DC-DC converters, LIDAR systems, reaction wheels, and ion thrusters [6].

Previous research works have extensively studied the performance of various semiconductor devices, including Si MOSFETs, SiC MOSFETs, Si IGBTs, and GaN devices at cryogenic temperatures. For example, Chailloux *et al.* and Chen *et al.* [7], [8] investigated the on-resistance and switching time changes of SiC MOSFETs under cryogenic conditions. Their findings indicate a significant increase in R_{ds-on} and a noticeable decrease in switching speed at lower temperatures. Most studies have focused on characterizing devices down to 77K using liquid nitrogen (LN₂) [9]–[17]. Nela *et al.* [18] performed the characterization of GaN devices down to 4.2K. However, they used a single sample for each part number, with four part numbers for the characterization. This approach does not accurately depict the behavior of the entire family from a single device characterization.

Given the lunar surface's wide temperature range—spanning from 100K to nearly 400K during lunar days and between 50K and 100K during lunar nights [19]–[21]—it is crucial to ensure that both the on-state performance and the off-state block-

ing/leakage current performance of devices are adequate across this spectrum (50K to 400K). Certain space applications, such as lunar missions, require operation at temperatures as low as 50K. The existing research on the static characterization of GaN devices is insufficient to fully understand and optimize their performance under these extreme conditions, as it has been limited to characterizations down to 77K. This paper addresses this gap by providing a comprehensive characterization of GaN devices across the full temperature range from 50K to 400K.

Moreover, the statistical variation in the static parameters at low temperatures poses a concern for emerging GaN technology. This is attributed to manufacturing inconsistencies and reliability immaturity compared to its Si counterpart. For example, a commercial GaN part exhibited a tolerance range of -35% to +53% in threshold voltage [22]. Such differences are critical, especially when devices are utilized in parallel or series configurations. Consequently, characterizing multiple GaN devices under consistent environmental conditions becomes necessary, necessitating the development of a multi-GaN static characterization platform. Ahmed et al. [23] discussed the automated characterization of multiple power devices, although their study was focused on Si IGBTs.

Characterizing multiple GaN devices across a wide temperature range (50K to 400K) is a lengthy and time-consuming process that traditionally requires continuous human monitoring. This includes tasks such as changing the temperature, starting the test, waiting for the test to complete, connecting the next sample, adjusting the temperature, waiting for the temperature to stabilize, saving the data, and processing the results. Although these tasks may seem numerous, much of the time is spent waiting—for the test to finish or for the temperature to change. To address the intensive need for human oversight, we aimed to eliminate the need for manual intervention by developing a fully automated platform that allows for the characterization of multiple devices with the press of a single button.

This paper focuses on the automated static characterization of multiple GaN devices using a cryocooler to achieve a broad temperature range. The goal is to enable testing with a single click of a button, eliminating the need for human intervention. This approach contrasts with other setups that rely on LN₂ cryochambers, which are limited to temperatures up to 77K.

II. CHARACTERIZATION AUTOMATION PLATFORM

A high-level view of our static automation characterization platform is shown in Fig. 1, and an overview of the actual hardware is provided in Fig. 4. The platform consists of a cryocooler, a programmable selector, a curve tracer, and a PC with a MATLAB GUI for automation. The cryocooler is a specialized refrigerator capable of achieving temperatures as low as 50K. It is used to cool the devices under test—multiple GaN devices placed inside the cryocooler for static characterization. The characterization process is conducted using a curve tracer. This device, a power/impedance analyzer, seamlessly combines the functionalities of a power source and

an oscilloscope for device characterization. Since only one GaN device can be characterized at a time, the programmable selector—comprising a relay array and IGBT switches—serves as an interface between the curve tracer and the multi-GaN board. This selector is controlled by a DSP card, allowing for automated switching between devices on the board. The entire system, including temperature control within the cryocooler, operation of the programmable selector, DSP card functions, curve tracer operations, and data management, is overseen by a Human-Machine Interface (HMI) unit. The HMI unit, which uses MATLAB for communication and automation, facilitates seamless operation and data processing.

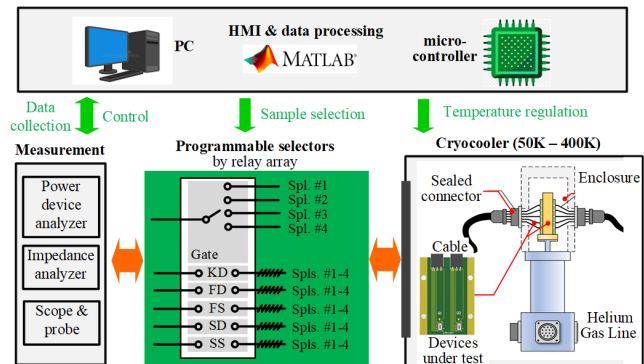


Fig. 1. Overview of the characterization automation platform

A. Cryocooler Setup

The CTI Cryogenics M22C Cryodone is employed to achieve cryogenic temperatures, serving as a cooling system. This system encompasses a cold head, a compressor unit, a vacuum pump, and a temperature controller. The cooling process of a cryocooler is based on a Gifford-McMahon (GM) refrigeration cycle, which leverages helium gas as the working fluid to approach absolute zero temperature [24]. Within the cold head, a cold station is present where the load requiring cooling is affixed. In our design, we mount the GaN board, comprising three GaN devices, onto the copper head. This copper head is then connected to the cold station, as illustrated in Fig. 2. Two GaN boards are fixed on the copper head, surrounded by the radiation shield to prevent radiation from adding additional load to the cryocooler, and finally covered by the enclosure. The enclosure has six holes for placing the electrical connectors. Maintaining a proper vacuum inside the cold head is vital for achieving cryogenic temperatures. The vacuum is created with a vacuum pump and measured by a vacuum meter. Given that the enclosure has six holes for placing vacuum connectors, any imperfect sealing during the placement of connectors can cause a significant temperature difference. Therefore, we used a cleanroom swab to clean the surface of the connectors and the O-ring, applied vacuum grease in the slot where the O-ring is placed, and then sealed the connectors.

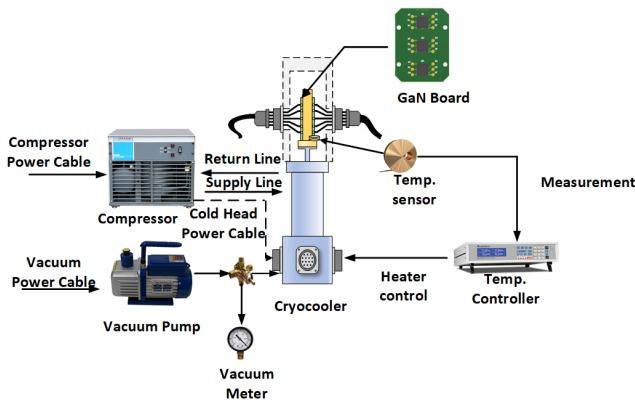


Fig. 2. Cryocooler setup for achieving cryogenic conditions

The radiation shield plays a pivotal role in attaining the minimum temperature. We observed a substantial difference in the minimum temperature the cryocooler could reach with and without the radiation shield. Without the radiation shield, the temperature plateaued around 117K and did not descend below this point. However, with the radiation shield in place, the temperature was reduced, reaching a low of 16K, as illustrated in Fig. 5. This is a significant improvement, even though our target temperature was only 50K.

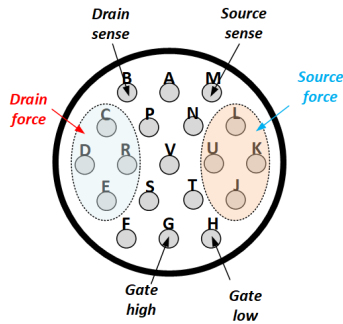


Fig. 3. Pin assignments in the hermetic connector for force, sense, and gate pairs

Initially, the goal was to establish a platform for characterizing six devices. As a result, all design and optimization efforts were focused on accommodating these six devices. For each device, six leads are required for characterization, totaling 36 leads, which contributes significantly to the thermal load on the cryocooler. The selection of appropriate leads inside the cryocooler is crucial for achieving temperatures as low as 50K. If the leads are poorly designed, they can introduce heat and increase the thermal load, hindering the achievement of the desired temperature [25], [26]. Although increasing the length of the leads and decreasing their cross-sectional area can reduce the thermal load, it also increases the resistance of the leads, potentially reducing the available voltage for characterization. To address this challenge, the lead design was optimized using four AWG 28 copper leads for the force drain

and force source cables, and AWG 32 for the gate and sense cables inside the cryocooler.

TABLE I
WIRE AWG AND MATERIAL SPECIFICATIONS FOR FORCE, SENSE, AND GATE PAIRS, AND TEMPERATURE SENSORS

Type	Force Pairs	Sense Pairs	Gate Pairs	Temperature Sensor
Current	High (Pulse Current)	Low (Pulse Current)	Low (Pulse Current)	Low
AWG	28	32	32	32
Material	Copper	Phosphor Bronze	Phosphor Bronze	Phosphor Bronze

Within the enclosure, two temperature sensors are installed: one on the cold station and the other on the GaN board. These sensors are connected to a temperature controller for temperature monitoring. Ensuring optimal thermal contact between the cold station and the copper header, as well as between the copper header and the GaN board, is critical. For this purpose, indium, a suitable thermal interface material for cryogenic applications, was employed. Without using indium, a significant temperature difference between the two sensors was observed. However, after applying indium, the temperature difference between the two sensors was reduced to around 2K, even at the lowest temperature of 16K, as illustrated in Fig. 5.

To maintain the temperature at a specific value, a temperature controller (LTC-11 from Neocera) was implemented. This controller not only monitors the temperature but also provides a heating function. The temperature controller is connected to the temperature sensor using a Kelvin connection with four wires: V+, I+, V-, and I-, as depicted in Fig. 6. The temperature controller supplies a positive current of 10 μ A, and the corresponding voltage drop is measured. This voltage is converted into a temperature value using an ADC and data conversion. Additionally, the controller can be remotely controlled via serial communication using an RS232 to USB cable.

B. Static Characterization

The static characterization platform is designed to perform three primary characterizations using the curve tracer: V-I characterization, reverse conduction, and breakdown/leakage voltage testing.

Moreover, various static parameters such as on-state resistance, threshold voltage, and transfer conductance can be derived through data processing. For static characterization, three pairs of wires—Force Drain/Force Source (FD/FS), Sense Drain/Sense Source (SD/SS), and Gate/Gate Return (GH/GL)—extend from the curve tracer and are routed to the GaN devices via the programmable selector.

When creating a specific pair of cables, for instance, the GH/GL pair, attention must be paid to the wire construction. The wires should be properly twisted, as depicted in Fig. 7.

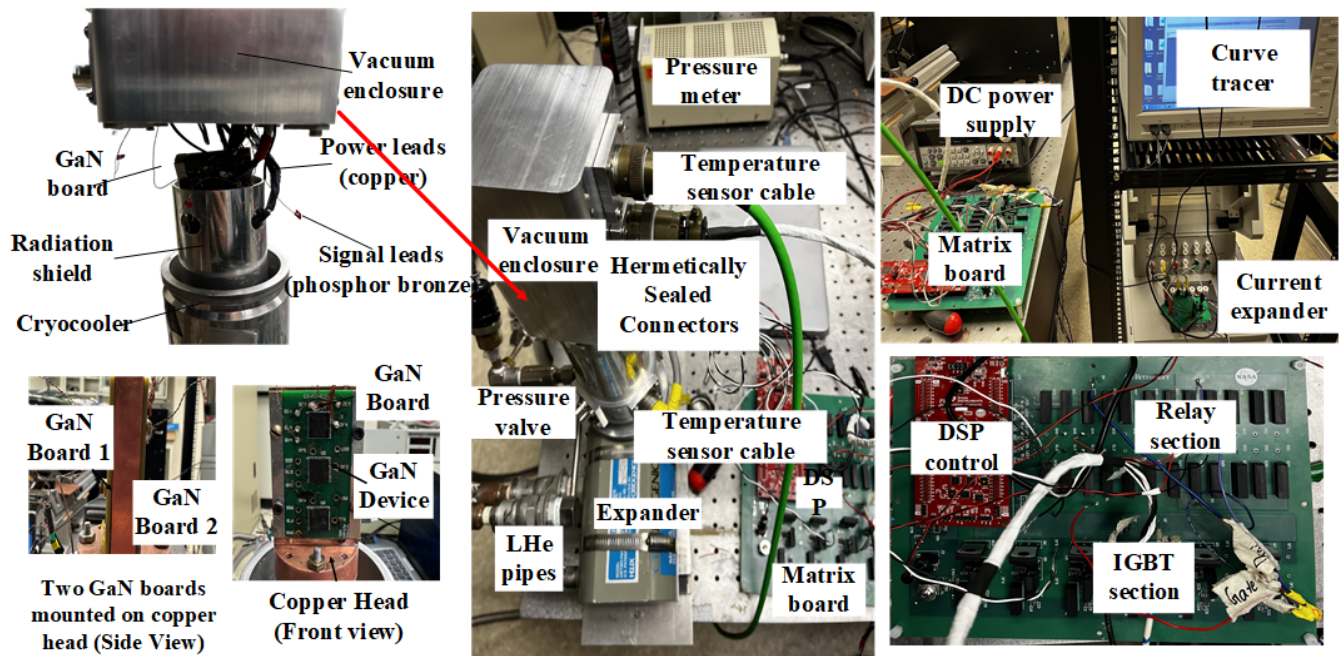


Fig. 4. (a) Top left: Cold head of the cryocooler showing the power and signal leads coming out of the radiation shield to the enclosure. (b) Bottom left: Side and front view of copper head with two GaN boards mounted. (c) Middle: Overview of the static characterization automation platform depicting the cold head, matrix board, vacuum enclosure, and pressure meter. (d) Top right: Illustration of the DC power supply, matrix board, and curve tracer connections. (e) Bottom right: Enlarged view of the matrix board depicting relays, IGBT switches, along with the DSP Card.

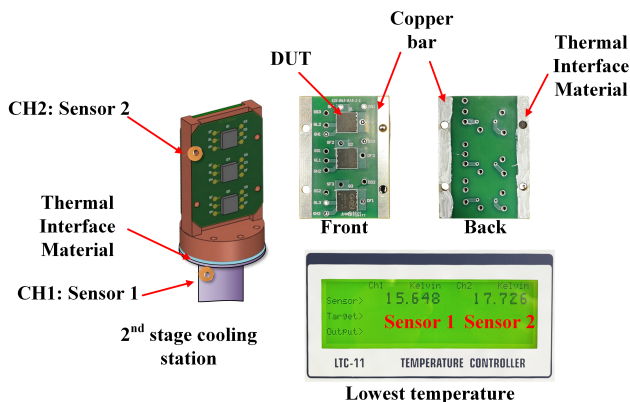


Fig. 5. (a) Left: Cold head showing the locations of temperature sensors (Sensor 1 and Sensor 2) and the GaN board. (b) Right: Top right shows the GaN board with front and back sides; the front side has DUTs soldered onto it. (c) Bottom right: Temperature controller displaying the minimum temperature achieved in the cryocooler, with readings from Sensor 1 and Sensor 2.

The loop area formed by the twisted wires should be minimized to reduce the transmission of EMI into that particular pair; otherwise, it would hinder the acquisition of accurate characterization results.

C. Programmable Selector

The curve tracer is designed to characterize only one device at a time. However, given multiple devices within the cryocooler, a programmable selector switch is essential

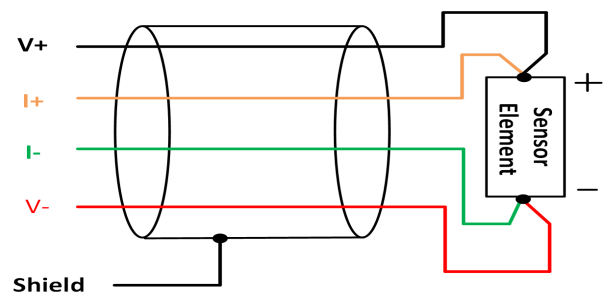


Fig. 6. Sensor cable connections showing V+, V-, I+, and I- wires

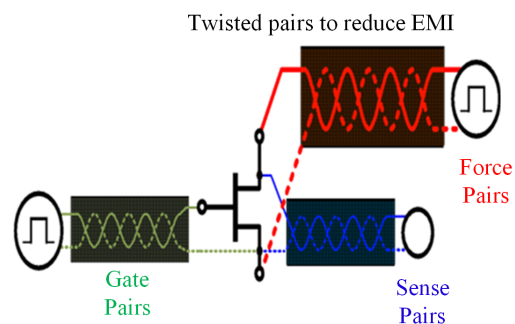


Fig. 7. Twisted pairs used for gate, sense, and force signals to reduce EMI

to select and connect exclusively to a specific device for characterization. For the Sense (SD/SS) and Gate (GH/GL) pairs, we utilize relays, as they carry negligible current. On the other hand, the Force pairs, which are required to withstand high voltages, carry substantial currents. High voltage/high current relays are both bulky and costly, prompting us to choose Insulated Gate Bipolar Transistors (IGBTs) as a more compact and cost-effective solution.

In fact, we employ a four-quadrant switch composed of IGBTs. Two anti-series combinations of IGBTs (each IGBT also includes an anti-parallel diode within the same package) constitute a four-quadrant switch, serving as an electrical switch for Drain Force (DF), as depicted in Fig. 8. For forward conduction, the current flows through the IGBT of the first switch and the diode of the second switch. For reverse conduction, the current flows through the diode of the first switch and the IGBT of the second switch. Therefore, for both forward and reverse conduction, either the IGBT or the diode conducts for each switch. The four-quadrant switch (anti-series combination of two IGBTs) is necessary because without it, reverse characterization of a specific device would not be possible, as using an IGBT with a diode alone could not block the negative voltage in the remaining devices that are not being characterized.

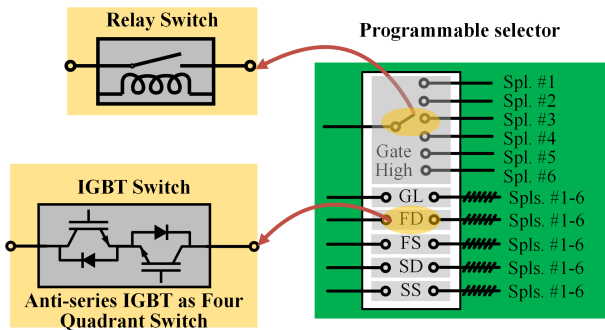


Fig. 8. Relays and IGBTs used as programmable selector switches

III. MATLAB GUI AND AUTOMATION

The primary feature of this static characterization platform is its automation capability. This feature enables the characterization of all devices across various temperatures with a single click facilitated by communication protocols. A MATLAB GUI has been developed to ensure easy and seamless automation by enabling efficient communication between MATLAB, the DSP card, the curve tracer, and the temperature controller.

Although we have developed the characterization platform for six devices, due to space limitations for cable placement inside the enclosure, we have limited the characterization to only four devices. Fig. 9 illustrates the flowchart for the automation process. Initially, communication is established between MATLAB and the Temperature Controller, the Matrix Board, and the Curve Tracer. Once this setup is complete, we select the starting temperature, the starting device, and the starting test. The system then performs the selected test

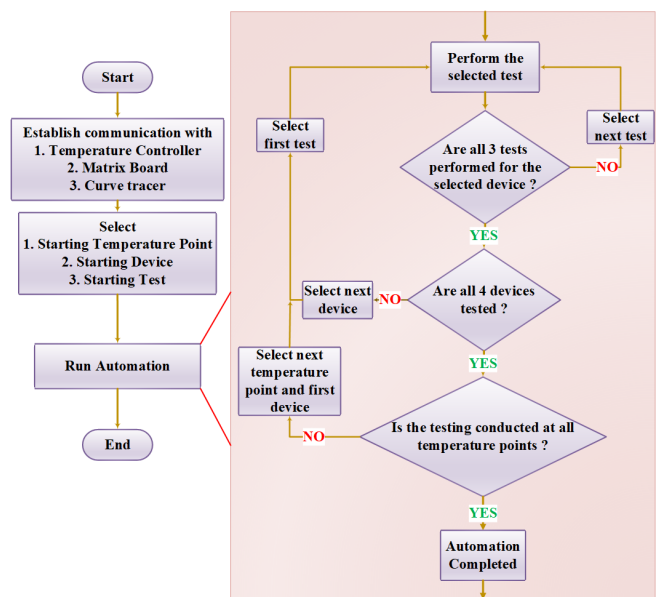


Fig. 9. Flowchart illustrating the implementation of automation in the MATLAB GUI

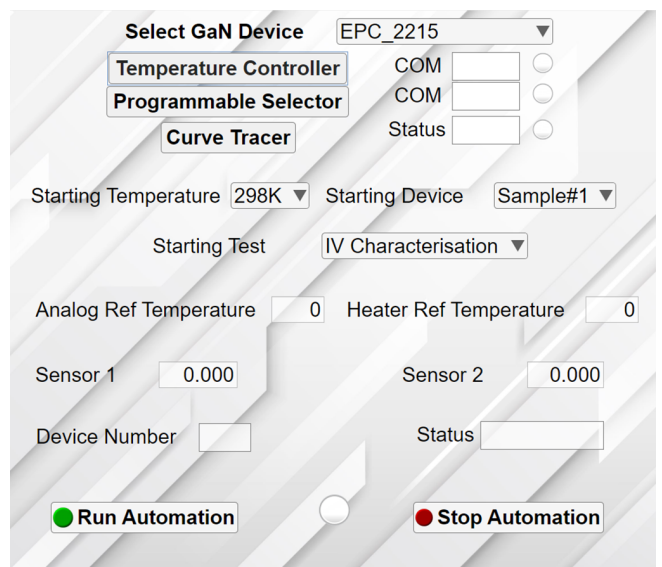


Fig. 10. A MATLAB GUI for integrating automation into the static characterization platform

and checks if all three tests—forward conduction, reverse conduction, and breakdown/leakage current—are completed. If not, it selects the next test. However, if all tests are completed, it checks if all four devices have been characterized. If not, it selects the next device for characterization and performs all three tests at that particular temperature. If all devices have been characterized, it checks if the tests have been conducted for all the predefined temperature points. If not, it selects the next predefined temperature point and performs all three characterizations, starting from the first device and moving to all four. If all temperature points have been tested, the

automation process is completed.

The total temperature points for characterization are 398K, 373K, 348K, 323K, 298K, 273K, 243K, 213K, 183K, 153K, 123K, 93K, 63K, and 50K, making a total of 14 temperature points. The characterization is performed twice for the intermediate temperature ranges, except for 50K and 398K. This ensures a comprehensive and accurate characterization of the devices.

IV. EXPERIMENTAL RESULTS

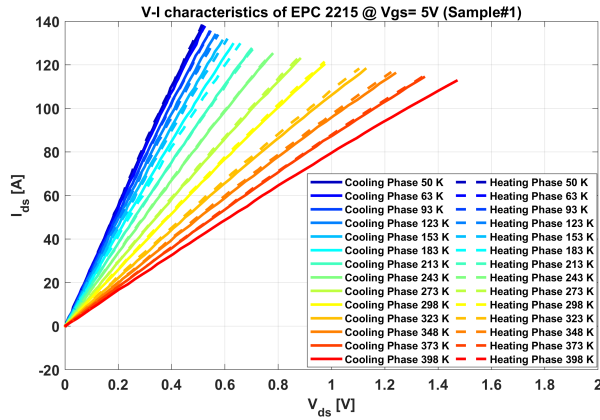


Fig. 11. V-I characteristics of EPC 2215 at $V_{gs} = 5V$ for Sample 1

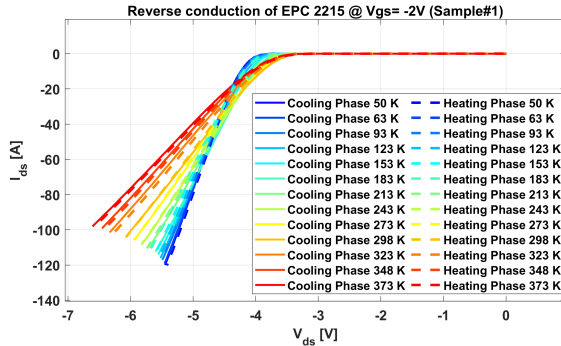


Fig. 12. Reverse conduction characteristics of EPC 2215 at $V_{gs} = -2V$ for Sample 1

Although the test platform was designed to characterize six devices within the cryocooler, the characterization could only be performed for four samples. This limitation was due to the confined space inside the enclosure, which did not allow the placement of wires for the other two devices.

Four samples from EPC System, specifically EPC 2215 (200V, 32A, $8m\Omega$) [27], were used to test the static characterization platform. During the experiment, one of the EPC devices (Sample 3) was damaged, limiting our data collection to only three devices. With our lead design and a radiation shield, we minimized the thermal load and successfully performed the characterization down to 50K.

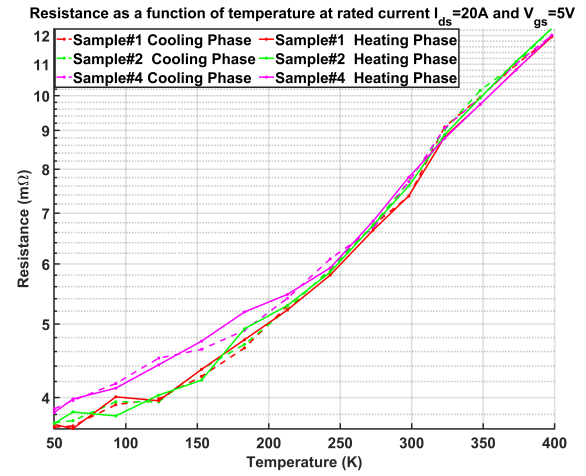


Fig. 13. On-state resistance versus temperature for EPC 2215 Samples 1, 2, and 4 at $I_{ds} = 20A$ and $V_{gs} = 5V$

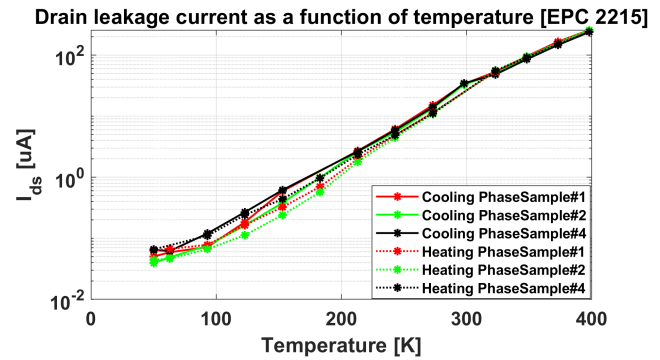


Fig. 14. Leakage current versus temperature for EPC 2215 Samples 1, 2, and 4

The I-V characterization results obtained from our test platform are shown in Fig. 11. These results reveal that the slope of the linear region increases at lower temperatures, indicating a decreasing trend in the on-state resistance. This trend is further verified in Fig. 13. Notably, the resistance at 50K is approximately two times lower than that at ambient temperature for the rated current, indicating more efficient conduction at lower temperatures.

The variation in on-state resistance between samples 1, 2, and 4 under an identical cryogenic environment from the same vendor and with the same part number is also observed in Fig. 13. Furthermore, the variation in on-state resistance of the sample devices during the heating and cooling phases in the thermal cycling is noted. For this particular part from EPC, considering both the heating and cooling phases for three samples, the minimum value observed at 123K is $3.94 m\Omega$, and the maximum value observed is $4.51 m\Omega$. While this variation may appear minimal, it might not be true for other part numbers from the same vendor or other vendors, and this variation might increase as the devices age and undergo numerous thermal and power cycles.

For reverse characterization at $V_{gs} = -2V$, we observed the diode-like behavior of the GaN device, as depicted in Fig. 12. For this specific part (EPC 2215), there was a certain variation in the knee voltage, with the knee voltage being lower at lower temperatures. Similar to forward conduction, the slope of the linear region at higher currents increases with decreasing temperature, indicating reduced on-state resistance and, in general, more efficient reverse conduction.

Additionally, the leakage current, as illustrated in Fig. 14 exhibited a decreasing trend as the temperature decreased from 398K to 50K, being minimal at 50K, indicating the robust voltage-blocking capability of the device during the off-state.

V. CONCLUSIONS AND FUTURE WORK

In this paper, we underscore the importance of developing a multi-GaN static characterization platform to ensure consistent testing conditions. The key features of this platform include its ability to perform characterization down to 50K and its automation capabilities. We presented the static characterization results for three samples of EPC 2215.

The static characterization of the EPC 2215 devices demonstrated a consistent decrease in on-state resistance as the temperature decreased from 398K to 50K. This indicates more efficient conduction under both forward and reverse conditions. Furthermore, decreased leakage current at lower temperatures suggests an improved blocking voltage capability.

Variations in on-state resistance among samples from the same vendor underscore the necessity of characterizing multiple GaN devices. These variations affect the operational tolerance of GaN devices. Moreover, the failure of one device during automated testing raises concerns about the reliability of these devices, reflecting the immaturity of this manufacturing technology.

In future work, we plan to consider more devices with similar voltage ratings and on-state resistance but with different packages to observe static parameter variations among the devices. Ultimately, this data, along with the observed static variation, could be leveraged with dynamic characterization results to create a comprehensive model of a GaN device. This model would cover the behavior of GaN devices over a wide range, making it suitable to simulate the operation of a power converter used in lunar missions.

ACKNOWLEDGMENT

This material is based on work supported by NASA under Award No. 80NSSC24K0179. The authors would also like to thank Mr. Richard C. Oeftering, Dr. Nicholas R. Uguccini, Dr. Brian A. Holler, and Mr. Frederick W. Van Keuls from NASA Glenn Research Center for their contributions to the design of the cryocooler platform. Additionally, we extend our gratitude to Prof. Ishwara Bhat from Rensselaer Polytechnic Institute for his assistance with the operation and maintenance of the cryogenic apparatus.

REFERENCES

- [1] EPC Space, "Space Level Qualified 40 to 300 V Rad Hard eGaN Power Transistors," [Online].
- [2] M. Zafrani, "Radiation Performance of Enhancement-Mode GaN Power Devices," [Online].
- [3] "Renesas' Rad-Hard 100V and 200V GaN FET DC-DC Solutions Use EPC Die," [Online].
- [4] ABB, "Failure Rates of IGBT Modules Due to Cosmic Rays," [Online].
- [5] H. Mohamed and D. Sanchez, "Key Parameters and Driving Requirements of GaN FETs," Texas Instruments, Application Brief SLUAAM0, Aug. 2022. [Online].
- [6] M. Zafrani, "Radiation Resilient and Robust: GaN Tech Propels Next-Gen Power Electronics to New Frontiers," EE Power, Feb. 15, 2024. [Online]. Available: <https://eepower.com/technical-articles/gan-technology-exhibits-extreme-robustness-for-space-missions/>. [Accessed: 19-Jul-2024].
- [7] T. Chailloux, C. Calvez, N. Thierry-Jebali, D. Planson, and D. Tournier, "SiC Power Devices Operation from Cryogenic to High Temperature: Investigation of Various 1.2 kV SiC Power Devices," *Materials Science Forum*, vol. 600-603, pp. 1122–1125, 2009.
- [8] S. Chen, C. Cai, T. Wang, Q. Guo, and K. Sheng, "Cryogenic and High Temperature Performance of 4H-SiC," in *Proc. IEEE Applied Power Electronics Conference and Exposition*, Long Beach, CA, 2013, pp. 207–210.
- [9] Z. Zhang, "Characterization of Wide Bandgap Device for Cryogenically-Cooled Power Electronics in Aircraft Applications," in *AIAA/IEEE EATS*, 2018.
- [10] M. C. Gonzalez, L. W. Kohlman, and A. J. Trunek, "Cryogenic Parametric Characterization of Gallium Nitride Switches," Glenn Research Center, Cleveland, Ohio, NASA/TP-2018-219973, 2018.
- [11] J. Colmenares *et al.*, "Experimental Characterization of Enhancement Mode Gallium-Nitride Power Field-Effect Transistors at Cryogenic Temperatures," in *IEEE WIPDA*, 2015, pp. 129–134.
- [12] F. Wang, Z. Zhang, and E. Jones, "Characterization of Wide Bandgap Power Semiconductor Devices," *Institution of Engineering and Technology*, Sept. 2018.
- [13] R. Ren *et al.*, "Characterization of 650 V Enhancement GaN HEMT Under Cryogenic Temperature," in *Proc. IEEE Energy Conversion Congress and Exposition*, Sept. 2018, pp. 891–897.
- [14] M. Mehrabankhomartash *et al.*, "Static and Dynamic Characterization of 650 V GaN E-HEMTs in Room and Cryogenic Environments," in *Proc. IEEE ECCE*, 2021, pp. 5289–5296.
- [15] A. M. E. Awwad, A. M. Al-Quteimat, and M. M. Al-Soud, "Static and Dynamic Characterization of GaN HEMT Power Semiconductor Devices," *Tafila Technical University*, 2024.
- [16] Z. Zhang *et al.*, "Characterization of High-Voltage High-Speed Switching Power Semiconductors for High Frequency Cryogenically-Cooled Application," in *Proc. IEEE APEC*, 2017, pp. 1964–1969.
- [17] H. Gui *et al.*, "Review of Power Electronics Components at Cryogenic Temperature," *IEEE Transactions on Power Electronics*, vol. 35, no. 5, May 2020, pp. 5144–5156.
- [18] L. Nela, N. Perera, C. Erine, and E. Matioli, "Performance of GaN Power Devices for Cryogenic Applications Down to 4.2 K," *IEEE Transactions on Power Electronics*, vol. 36, no. 7, 2021, pp. 7412–7416.
- [19] NASA, "Space Technology Research Grants Program, Early Career Faculty Appendix, NASA Research Announcement (NRA): Space Technology – Research, Development, Demonstration, and Infusion 2022 (SpaceTech-REDDI-2022)," 80HQTR22NOA01.
- [20] R. C. Oeftering, "Lunar Power Hibernation for Surviving the Lunar Night," presented at Lunar Extreme Environments Forum, 2021. [Online]. Available: <https://ntrs.nasa.gov/citations/20210019184>. [Accessed: 19-Jul-2024].
- [21] NASA Glenn, "Contributions to the International Space Station (ISS) Electrical Power System," [Online]. Available: <https://www.nasa.gov/centers/glenn/about/fs08grc.html>. [Accessed: 19-Jul-2024].
- [22] GaN Systems, "GS66516T Datasheet," [Online]. Available: <https://www.gansystems.com/datasheets/gs66516t-datasheet.pdf>. [Accessed: 19-Jul-2024].
- [23] A. Siraj *et al.*, "Integrated AC Power Cycling Platform with Automated Characterization for T-Type Power Module in Photovoltaic Application," in *Proc. IEEE ECCE*, Nashville, TN, 2023.

- [24] CTI Cryogenics, "Cryodyne Refrigeration System Models 22, 350, 1020, and 1050," Datasheet, [Online].
- [25] R. McFee, "Optimum input leads for cryogenic apparatus," *Review of Scientific Instruments*, vol. 30, no. 2, pp. 98-102, 1959.
- [26] J. Ekin, *Experimental Techniques for Low-Temperature Measurements: Cryostat Design, Material Properties and Superconductor Critical-Current Testing*. Oxford, UK: Oxford University Press, 2006.
- [27] Efficient Power Conversion Corporation, "EPC2215 eGaN® FET Datasheet," June 2023. [Online].